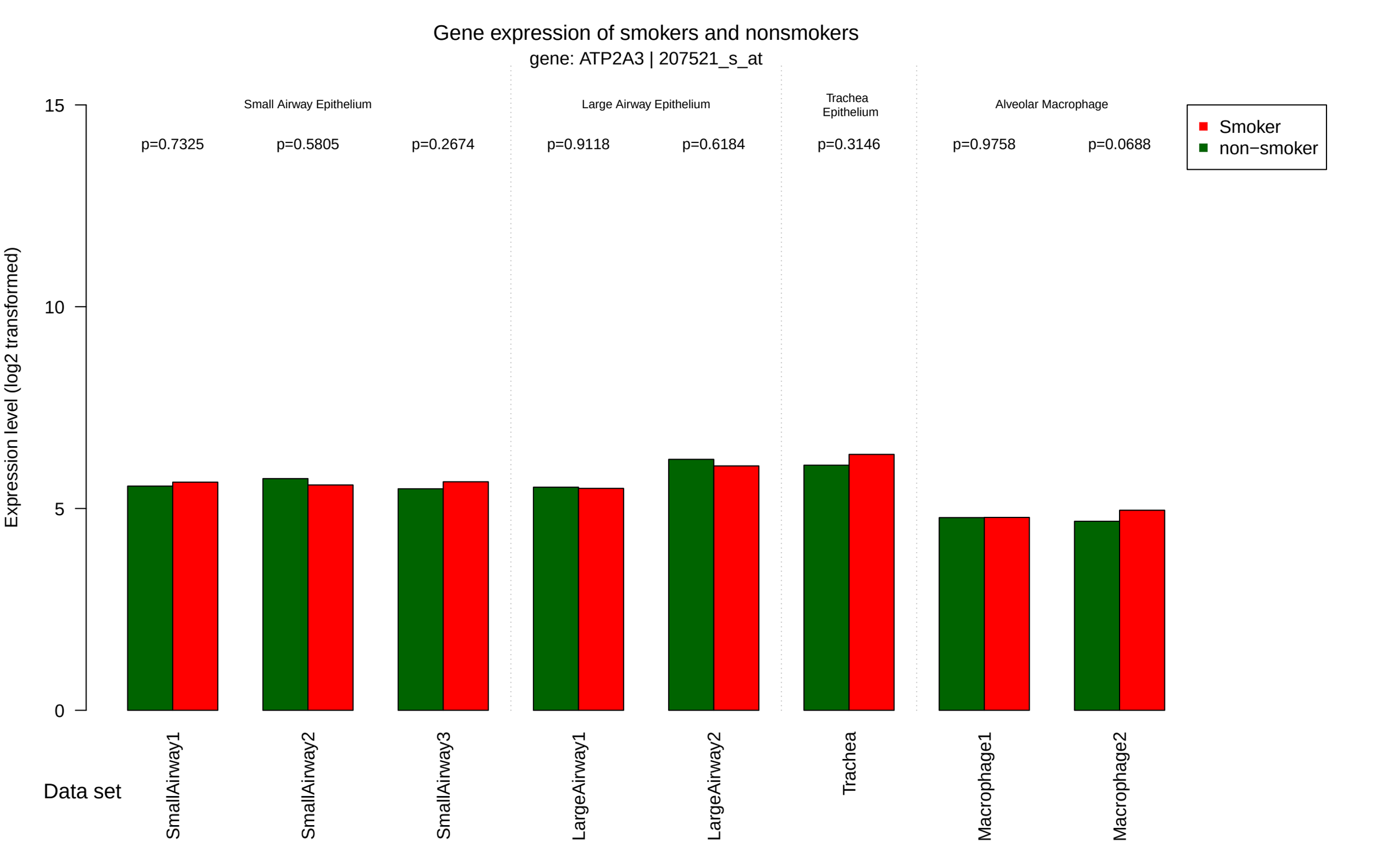
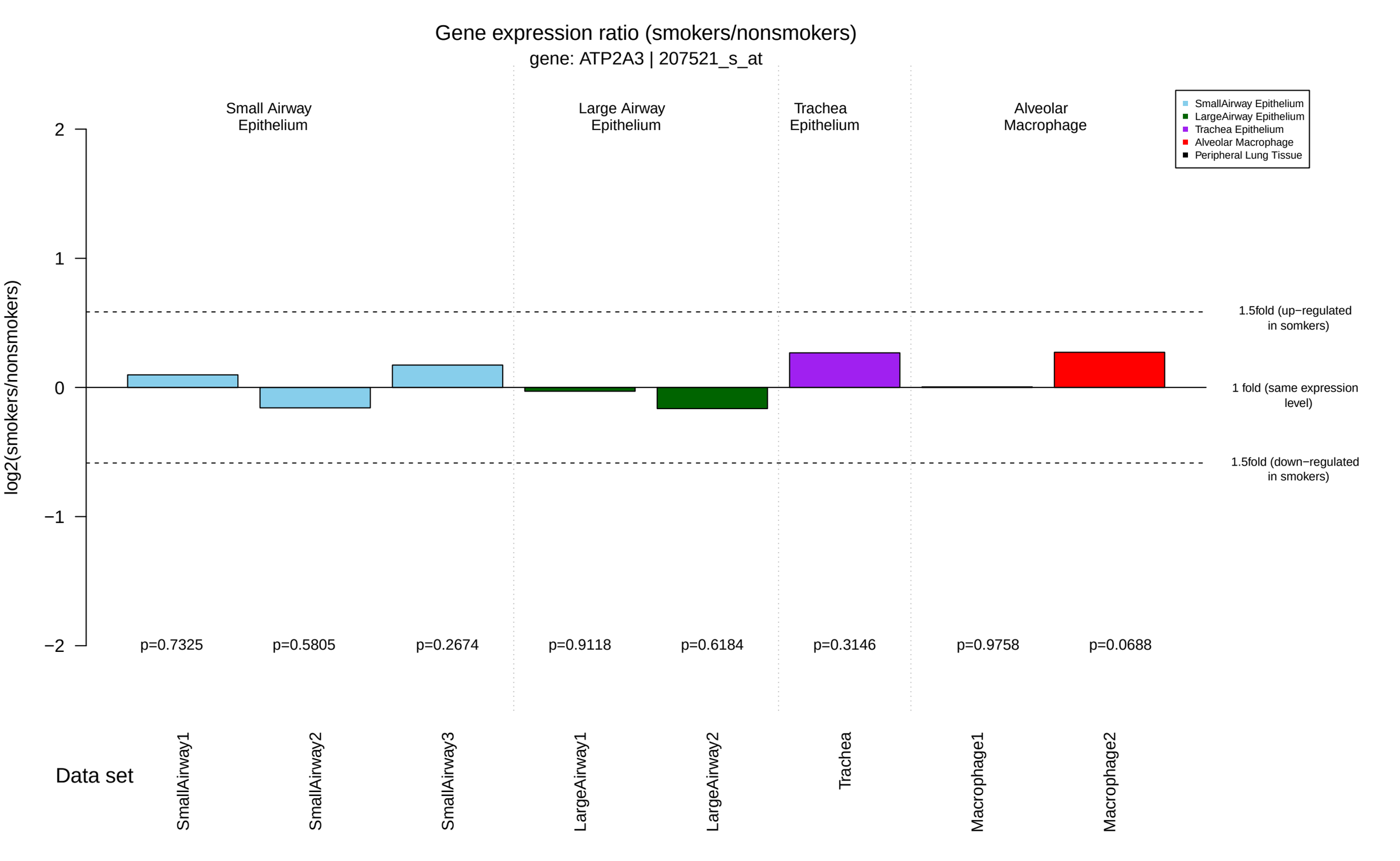
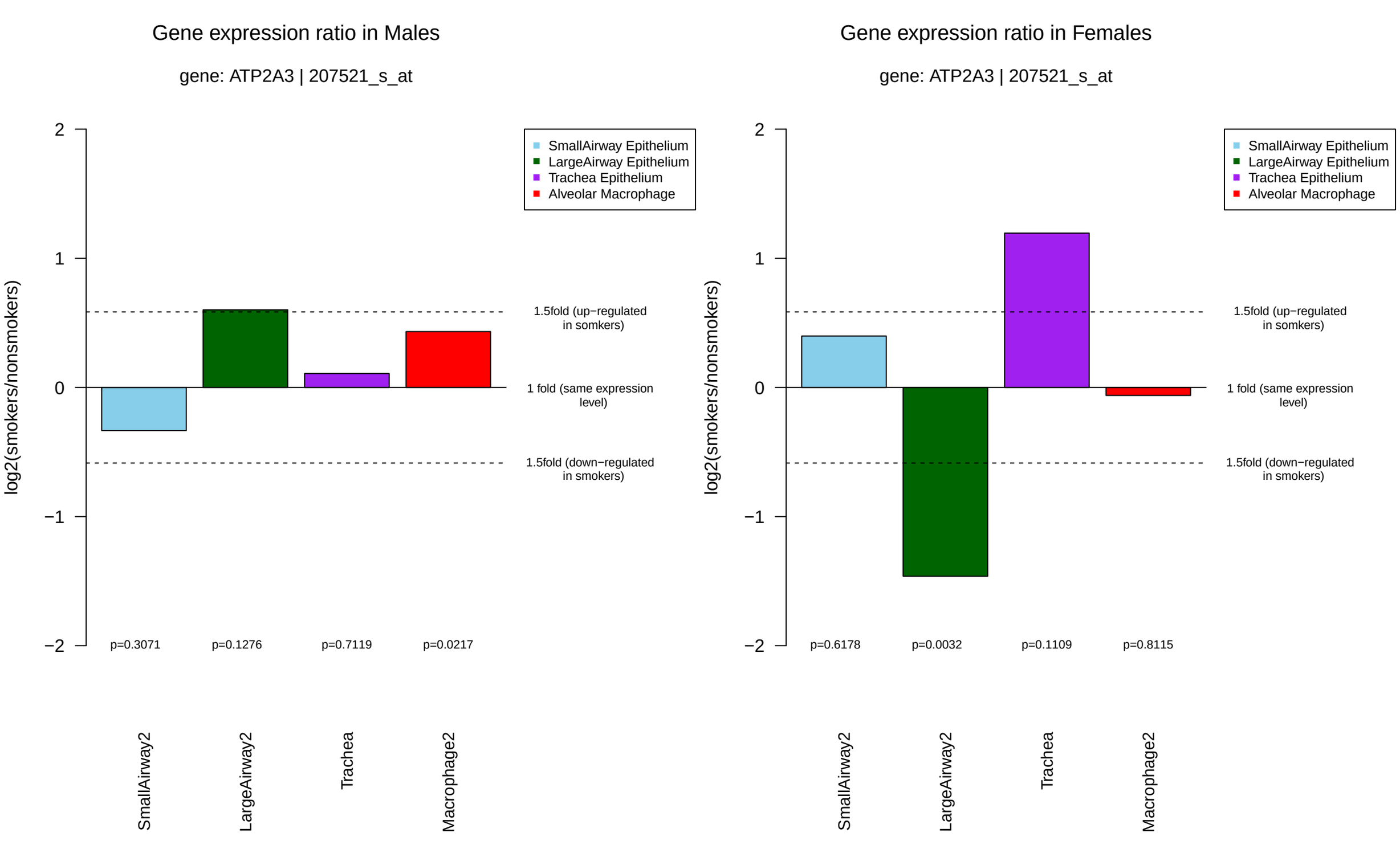
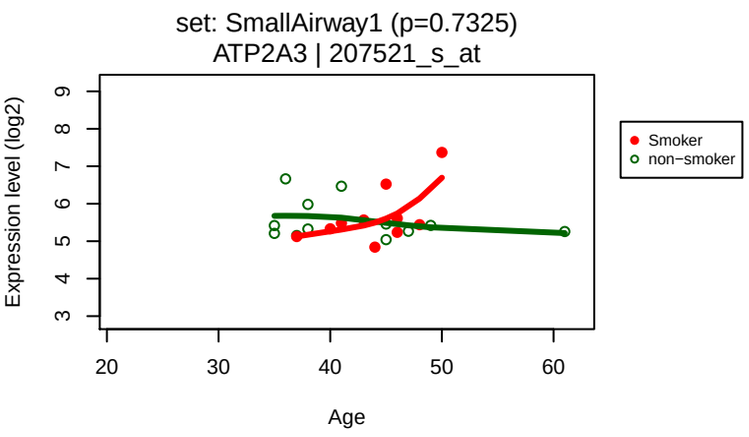
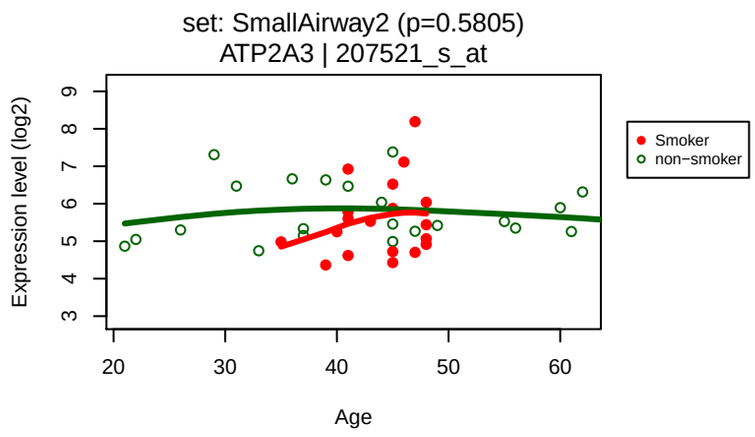
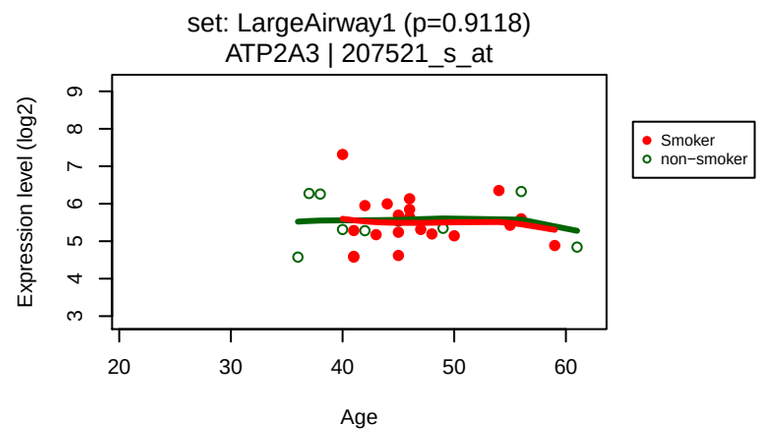

Gene expression of smokers and nonsmokers
gene: ATP2A3 | 207521_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7325
p=0.5805
p=0.2674
p=0.9118
p=0.6184
p=0.3146
p=0.9758
p=0.0688
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

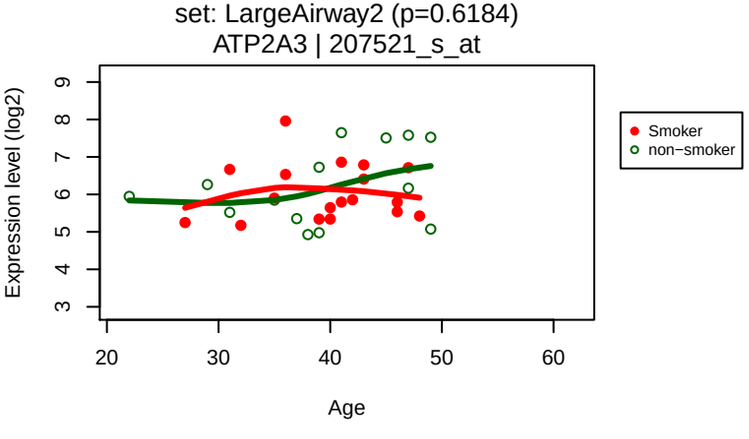

Gene expression ratio (smokers/nonsmokers)
gene: ATP2A3 | 207521_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7325
p=0.5805
p=0.2674
p=0.9118
p=0.6184
p=0.3146
p=0.9758
p=0.0688
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

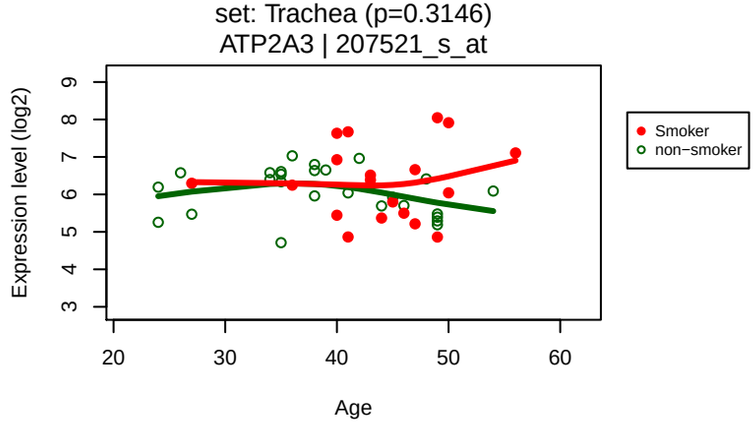

Gene expression ratio in Males
gene: ATP2A3 | 207521_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3071
p=0.1276
p=0.7119
p=0.0217
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2A3 | 207521_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6178
p=0.0032
p=0.1109
p=0.8115
SmallAirway2
LargeAirway2
Trachea
Macrophage2

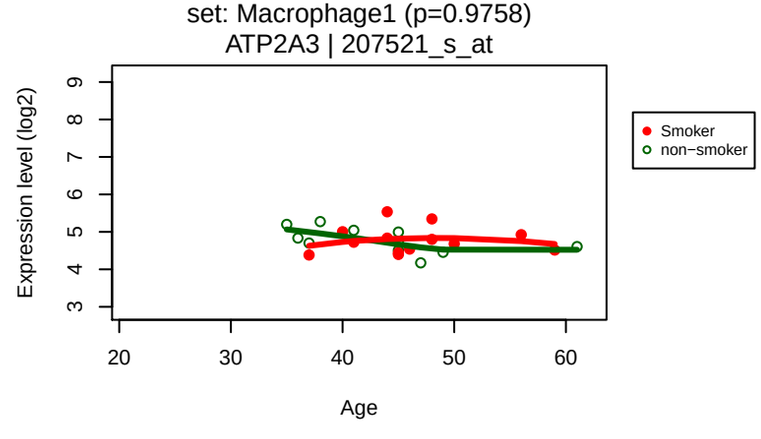

set: SmallAirway1 (p=0.7325)
ATP2A3 | 207521_s_at
Expression level (log2)
Age
Smoker
non−smoker

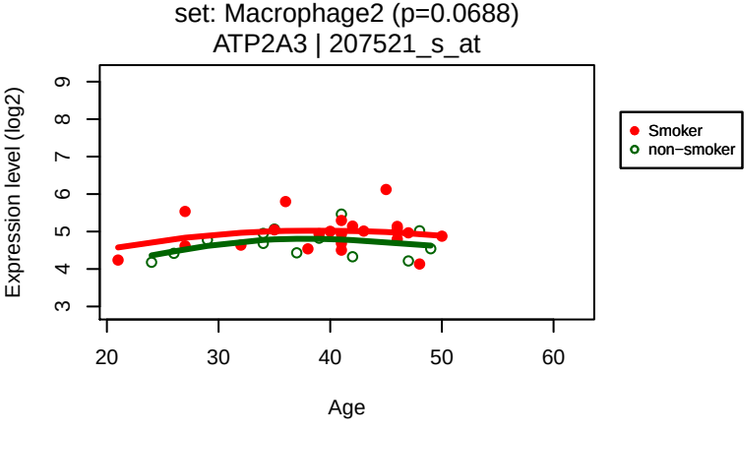

set: SmallAirway2 (p=0.5805)
ATP2A3 | 207521_s_at
Expression level (log2)
Age
Smoker
non−smoker

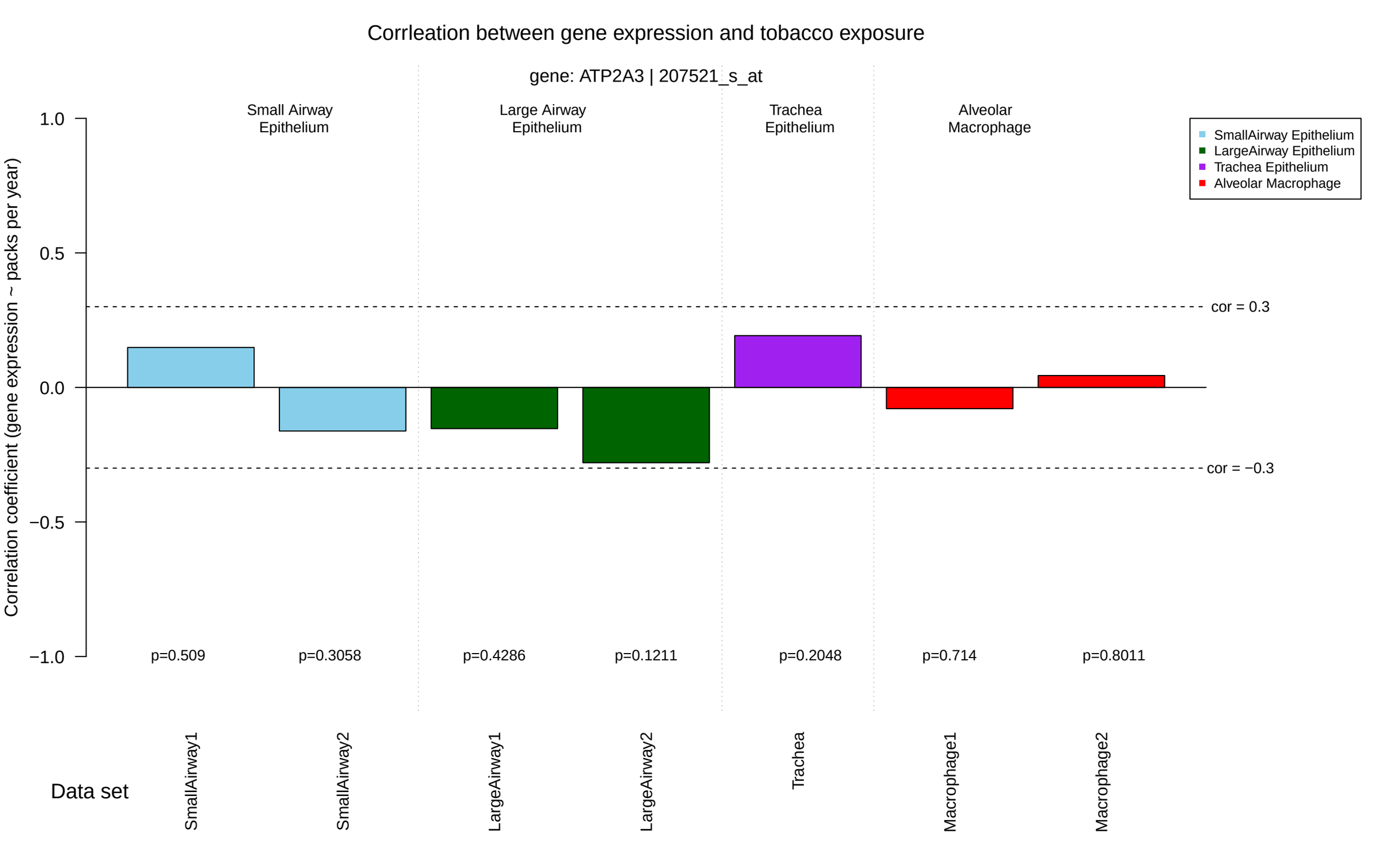

set: LargeAirway1 (p=0.9118)
ATP2A3 | 207521_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6184)
ATP2A3 | 207521_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3146)
ATP2A3 | 207521_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9758)
ATP2A3 | 207521_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0688)
ATP2A3 | 207521_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2A3 | 207521_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.509
p=0.3058
p=0.4286
p=0.1211
p=0.2048
p=0.714
p=0.8011
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2